

Assignment 2: Warehouse Storage Optimization Using Genetic Algorithm

Instructions:

- This is an individual Assignment. Demos will start soon after submission date; missing demo will result in straight zero marks.
- All parties involved in any kind of plagiarism/cheating will be given zero marks in all the assignments.
- Assignment deadline will not be extended.
- Late submissions will not be accepted, so submit and turn-in your assignment on-time.
- You must follow below submission guidelines, otherwise your submission won't be accepted:
 1. Create a new directory.
 2. Change its name to the following format YOURSECTION_ROLLNUMBER_NAME.
 3. E.g. A_20I-0012_Muhammad Ahmad
 4. Put all your files into this newly created directory.
 5. Compress the directory into a compressed .zip file.
 6. Submit it on Google Classroom.

Understanding the problem is also part of the assignment. As much as possible, the assignment has been explained in detail, alongside examples.

1. Objective

The objective of this assignment is to apply the Genetic Algorithm (GA) to solve a real-world optimization problem involving warehouse storage management. Students will design and implement a GA that determines the optimal placement of products in different warehouse storage zones while satisfying operational constraints.

2. Problem Overview

Modern warehouses handle thousands of products with different storage requirements. Efficient product placement is essential to improve space utilization, safety, retrieval speed, and operational efficiency.

The warehouse manager must decide where each product should be stored within the available storage zones. However, several physical and operational constraints must be satisfied.

Your task is to design a Genetic Algorithm that finds the optimal warehouse storage plan.

3. Warehouse Zones

The warehouse contains eight storage zones with different characteristics:

Zone	Description	Capacity (kg)
Z1	Heavy Item Floor Storage	120 kg
Z2	Standard Rack Storage	80 kg
Z3	Fragile Item Shelf	80 kg
Z4	Temperature Controlled Storage	80 kg
Z5	Hazardous Material Storage	80 kg
Z6	Fast-Moving Product Area (Near Exit)	60 kg
Z7	Bulk Dry Storage	150 kg

Z8	Refrigerated Loading Dock	100 kg
-----------	---------------------------	--------

4. Product Dataset

Each product has different characteristics. The dataset below must be used as-is. You may add additional products to further stress-test your algorithm, but all 20 base products are required.

Product	Weight	Category	Fragile	Hazardous	Temp Ctrl	Demand
P1 – Glass Bottles	20 kg	Beverage	Yes	No	No	Medium
P2 – Frozen Meat	30 kg	Food	No	No	Yes	High
P3 – Cleaning Acid	10 kg	Chemical	No	Yes	No	Low
P4 – Rice Bags	50 kg	Grocery	No	No	No	High
P5 – Ceramic Plates	15 kg	Kitchenware	Yes	No	No	Medium
P6 – Ice Cream	25 kg	Food	No	No	Yes	High
P7 – Detergent	12 kg	Chemical	No	Yes	No	Medium
P8 – Chips Carton	8 kg	Snacks	No	No	No	High
P9 – Olive Oil Bottles	18 kg	Grocery	Yes	No	No	Medium
P10 – Industrial Bleach	22 kg	Chemical	No	Yes	No	Low
P11 – Yogurt Cartons	14 kg	Food	No	No	Yes	High
P12 – Flour Bags	45 kg	Grocery	No	No	No	High
P13 – Wine Bottles	16 kg	Beverage	Yes	No	No	Medium
P14 – Paint Cans	28 kg	Chemical	No	Yes	No	Low
P15 – Biscuit Boxes	6 kg	Snacks	No	No	No	High
P16 – Motor Oil	35 kg	Chemical	No	Yes	No	Low
P17 – Frozen Fish	20 kg	Food	No	No	Yes	High
P18 – Bubble Wrap Rolls	10 kg	Packaging	Yes	No	No	Medium
P19 – Wheat Sacks	40 kg	Grocery	No	No	No	High
P20 – Hand Sanitizer	8 kg	Chemical	Yes	No	No	Medium

5. Constraints

Your Genetic Algorithm must handle all of the following constraints. Each violation contributes to the fitness penalty score.

5.1 Zone Weight Capacity

Each zone has a maximum weight limit. The total weight of all products assigned to a zone must not exceed its capacity. The penalty is applied proportionally to the excess weight.

Example: If Z2 capacity = 80 kg and total assigned weight = 100 kg, the excess is 20 kg, contributing +40 penalty points (20 kg x +2 per kg).

5.2 Fragile Product Protection

Fragile products must be placed in the Fragile Item Shelf (Z3). Placing fragile items in any other zone increases the risk of damage.

- Affected products: Glass Bottles, Ceramic Plates, Olive Oil Bottles, Wine Bottles, Bubble Wrap Rolls, Hand Sanitizer

5.3 Hazardous Material Isolation

Hazardous items must be stored exclusively in Hazardous Storage (Z5). They must not be placed near food or fragile items.

- Affected products: Cleaning Acid, Detergent, Industrial Bleach, Paint Cans, Motor Oil

5.4 Temperature-Controlled Storage

Products requiring refrigeration must be placed in Temperature Controlled Storage (Z4) or the Refrigerated Loading Dock (Z8). These products cannot be placed in standard racks or any other zone.

- Affected products: Frozen Meat, Ice Cream, Yogurt Cartons, Frozen Fish

5.5 Fast-Moving Product Accessibility

High-demand products should be placed in Z6 (near exit) to reduce picking time. Misplacement reduces operational efficiency.

- Affected products: Frozen Meat, Rice Bags, Chips Carton, Yogurt Cartons, Flour Bags, Biscuit Boxes, Frozen Fish, Wheat Sacks
- Note: Products with both a zone-specific constraint (e.g., temperature control) and a high-demand constraint receive penalties for each violation independently.

5.6 Heavy Item Placement

Items heavier than 40 kg must be placed in Floor Storage (Z1) to ensure forklift accessibility. Placing them in raised racks or fragile shelves is a violation.

- Affected products: Rice Bags (50 kg), Flour Bags (45 kg)

5.7 Product Compatibility

Products from the same category should ideally be stored in the same zone to improve picking efficiency. A penalty is applied for each pair of same-category products placed in different zones.

- Example: If P3 (Chemical) is in Z5 and P7 (Chemical) is in Z2, this counts as one separated pair and contributes +3 penalty.

5.8 Food and Chemical Incompatibility

Food products must never be stored in the same zone as Chemical products. This is a strict safety regulation. A penalty is applied for each Food-Chemical pair sharing a zone.

- Food products: Frozen Meat, Ice Cream, Yogurt Cartons, Frozen Fish

- Chemical products: Cleaning Acid, Detergent, Industrial Bleach, Paint Cans, Motor Oil, Hand Sanitizer
- Example: If P2 (Food) and P3 (Chemical) are both assigned to Z2, this is one violation contributing +15 penalty.

5.9 Loading Dock Restriction

The Refrigerated Loading Dock (Z8) is reserved exclusively for high-demand temperature-controlled products awaiting dispatch. Only products that are both temperature-controlled and high-demand may be assigned to Z8. Any other product placed in Z8 is a violation.

- Eligible products for Z8: Frozen Meat, Yogurt Cartons, Frozen Fish
- Note: Z8 is optional — eligible products may also be placed in Z4 without penalty, but non-eligible products in Z8 incur a penalty.

6. Fitness Function

The fitness function calculates the total penalty score for a given chromosome. A lower score is better. The goal is to minimize the total penalty to zero.

Fitness formula:

`fitness(chromosome) = sum of all penalties across all products and zones`

Penalty is applied per product that violates a constraint. Weight capacity penalty is applied per kg over the zone limit.

Constraint Violation	Penalty
Zone weight capacity exceeded	+2 per kg over limit
Fragile item placed outside Z3	+8 per product
Hazardous item placed outside Z5	+10 per product
Temperature-sensitive item placed outside Z4 or Z8	+9 per product
High-demand item not placed in Z6	+5 per product
Heavy item (>40 kg) not placed in Z1	+4 per product
Compatible items stored in different zones	+3 per separated pair
Food and Chemical items sharing a zone	+15 per pair
Non-eligible product placed in Z8	+12 per product

7. Genetic Algorithm Implementation

Your GA must include all six components described below.

7.1 Chromosome Representation

Each chromosome is a list of zone assignments, one per product. The index of each gene corresponds to a product (P1 to P20), and the value is the zone assigned to it (Z1 to Z8).

`Chromosome = [Z3, Z4, Z5, Z1, Z3, Z4, Z5, Z6, Z3, Z5, Z4, Z1, Z3, Z5, Z6, Z5, Z4, Z3, Z7, Z5]`

Each gene represents the zone assigned to the product at that index.

7.2 Initial Population

Generate an initial population of random warehouse storage plans. Population size should be configurable (suggested range: 50 to 200 individuals). Number of generations should also be configurable (suggested range: 100 to 500).

7.3 Selection Strategy

Implement one of the following selection methods. In your report, you must justify why you chose it and explain the tradeoffs compared to the alternatives.

- Roulette Wheel Selection
- Tournament Selection
- Rank Selection

7.4 Crossover

Implement a crossover operation to combine two parent chromosomes and produce offspring.

- Single-point crossover
- Two-point crossover

7.5 Mutation

Introduce mutation to maintain diversity. Example: randomly reassign the zone for one or more products in a chromosome with a small probability.

7.6 Termination Condition

Stop the algorithm when:

- The maximum number of generations is reached, or
- The best fitness score reaches 0 (all constraints satisfied).

8. Output Requirements

Your program must display the following:

- Best storage plan (which products are assigned to which zones)
- Best fitness score achieved
- Number of generations run
- List of any remaining constraint violations

Example output:

```
Optimal Storage Plan
Z1: Rice Bags, Flour Bags
Z2: (empty)
Z3: Glass Bottles, Ceramic Plates, Olive Oil Bottles, Wine Bottles, Bubble
Wrap Rolls, Hand Sanitizer
Z4: Frozen Meat, Ice Cream, Yogurt Cartons
Z5: Cleaning Acid, Detergent, Industrial Bleach, Paint Cans, Motor Oil
Z6: Chips Carton, Biscuit Boxes, Wheat Sacks
Z7: (empty)
Z8: Frozen Fish
```

9. Bonus Task (Extra Marks)

Store the optimized shelf allocation results in an Excel sheet for better visualization and analysis. The sheet should include all required columns.

10. What to Submit

- A single Python 3 file (e.g., assignment2.py).
- A word document that includes details on the following and your reasoning for your decisions:
 1. Chromosome representation chosen.
 2. Fitness function formula and how penalties are calculated.
 3. Selection strategy chosen and justification of why it was selected over the alternatives.
 4. Crossover method used.
 5. Mutation strategy.
 6. Justification for population size and generation count chosen.

Your code must run with a standard Python 3 interpreter. External libraries are permitted only for the bonus Excel export task.